MathDB 332617: a principal-log counterexample and the complete continuous-log formula

Principia Math In Progress

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Task 1: Full resolution and proof

The source is Conjecture 3.2 of Ariad, Segal, Avishai and Grosfeld, *Minimal multi-vortex configurations on a flat torus and cylinder*, arXiv:1603.00840v1, p. 9. Its notation (p. 8) is

$$z_m = \frac{x + mq_x + iy}{q_y}, \qquad x, y \in \mathbb{R}, \quad q_x, q_y > 0,$$

and it asks whether the sum and integral of

$$\operatorname{Log} R_A(m), \qquad R_A(m) = \frac{\Gamma(A+1+i\overline{z}_m)\Gamma(A+1-i\overline{z}_m)}{\Gamma(A+1+iz_m)\Gamma(A+1-iz_m)}$$

over $[-A, A]$ have equal limits as the integer $A \to \infty$. Appendix A.1 calls Log the principal value. We use its standard convention $\operatorname{Im}\operatorname{Log} \in (-\pi, \pi]$; the printed interval $(\pi, \pi]$ is an evident sign typo. We disprove the principal-value assertion by explicit unequal subsequential limits. We also prove the full all-parameter assertion with the continuous logarithm normalized at $\operatorname{Re} z = 0$, including its claimed common value. Thus the result does not consist solely of a branch-convention objection.

Write $a = A + 1$, $b = x/q_y$, $c = q_x/q_y > 0$, and $v = y/q_y$. Let ℓ be the holomorphic logarithm of Γ on the right half-plane which is real on the positive axis. For $a > |v|$ define

$$\begin{aligned} L_A(u) = {} & \ell(a + v + iu) + \ell(a - v - iu) \\ & - \ell(a - v + iu) - \ell(a + v - iu). \end{aligned} \tag{1}$$

Then $\exp L_A(b + cm) = R_A(m)$, and $L_A(0) = 0$.

Theorem 1 (Continuous logarithm, all parameters)**.** *For every fixed $b, v \in \mathbb{R}$ and $c > 0$, both limits exist and*

$$\lim_{A\to\infty} \sum_{m=-A}^{A} L_A(b + cm) = \lim_{A\to\infty} \int_{-A}^{A} L_A(b + cm)\, dm = \frac{8ibv}{c} \arctan c = \frac{8ixy}{q_x q_y} \arctan \frac{q_x}{q_y}. \tag{2}$$

For $B = A + 1 - |v| \geq 1$, the absolute difference between the finite sum and integral is at most

$$4|bv|(B^{-1} + B^{-2}) + Ac^2|v|(B^{-2} + 2B^{-3}) = O(A^{-1}). \tag{3}$$

Proof. Put $\psi = \ell'$. The usual locally uniformly convergent digamma series on $\operatorname{Re} w > 0$ is

$$\psi(w) = -\gamma + \sum_{n=0}^{\infty} \left(\frac{1}{n+1} - \frac{1}{n+w} \right). \tag{4}$$

It follows, for example, by logarithmic differentiation of Euler's limit product for Γ; subtract the harmonic sum before taking the limit. Local uniform convergence follows by bounding the paired terms by $O(n^{-2})$, and permits differentiation. Consequently

$$\psi'(w) = \sum_{n=0}^{\infty} (n + w)^{-2}, \qquad \psi''(w) = -2 \sum_{n=0}^{\infty} (n + w)^{-3}.$$

For $\operatorname{Re} w \geq B > 0$, comparison with the corresponding positive real integrals gives

$$|\psi'(w)| \leq B^{-2} + B^{-1}, \qquad |\psi''(w)| \leq 2B^{-3} + B^{-2}. \tag{5}$$

Conjugation symmetry of ℓ gives

$$L_A(u) = 2i \int_{-v}^{v} \operatorname{Im} \psi(a + t + iu)\, dt. \tag{6}$$

All integrals with reversed endpoints are oriented. In particular, L_A is odd in u and purely imaginary, including when $v < 0$. Differentiating twice and applying (5) yields the bounds

$$\sup_{u \in \mathbb{R}} |L_A'(u)| \leq 4|v|(B^{-1} + B^{-2}), \qquad \sup_{u \in \mathbb{R}} |L_A''(u)| \leq 4|v|(B^{-2} + 2B^{-3}). \tag{7}$$

For a twice continuously differentiable complex function f on an integer interval, two integrations by parts on each unit interval give the composite trapezoidal identity

$$T_A(f) - \int_{-A}^{A} f(t)\, dt = \frac{1}{2} \sum_{m=-A}^{A-1} \int_{m}^{m+1} (t - m)(m + 1 - t) f''(t)\, dt,$$

where T_A has half weights at the two endpoints. Its absolute value is at most $\frac{1}{8} \int_{-A}^{A} |f''(t)|\, dt$. Take $f(t) = L_A(b + ct)$. Oddness gives

$$f(A) + f(-A) = L_A(cA + b) - L_A(cA - b),$$

whose absolute value is at most $2|b| \sup |L_A'|$. The full sum differs from T_A by half this endpoint sum. Thus its error is bounded by $|b| \sup |L_A'| + (Ac^2/4) \sup |L_A''|$, proving (3).

It remains to evaluate the integral, without interchanging a nonuniform sum and a limit. Taking imaginary parts in (4), for $s > 0$ and real u, gives

$$\operatorname{Im} \psi(s + iu) = \sum_{n=0}^{\infty} \frac{u}{(s + n)^2 + u^2}.$$

For $u > 0$ the summand is decreasing in n. Its integral from zero to infinity is $\arctan(u/s)$, so

$$|\operatorname{Im} \psi(s + iu) - \arctan(u/s)| \leq \frac{|u|}{s^2 + u^2} \leq \frac{1}{2s}. \tag{8}$$

Oddness proves the same bound for $u \leq 0$. For bounded real h and $|t| \leq |v|$, it follows uniformly that

$$\operatorname{Im} \psi(a + t + i(cA + h)) \longrightarrow \arctan c, \qquad L_A(cA + h) \longrightarrow 4iv \arctan c.$$

Using oddness once again,

$$\int_{-A}^{A} L_A(b + cm)\, dm = \frac{1}{c} \int_{cA-b}^{cA+b} L_A(u)\, du \longrightarrow \frac{8ibv}{c} \arctan c.$$

This identity also holds for negative b by orientation and for $b = 0$. Together with (3) it proves (2). □

Theorem 2 (Explicit principal-value counterexample)**.** *Set $q_x = 2$, $q_y = 1$, $x = 1/2$, and $y = 1$. If*

$$S_A = \sum_{m=-A}^{A} \operatorname{Log} R_A(m), \qquad I_A = \int_{-A}^{A} \operatorname{Log} R_A(m)\, dm,$$

with principal Log*, then*

$$\lim_{n \to \infty} S_{2n} = 2i \arctan 2 - 2\pi i,$$
$$\lim_{n \to \infty} S_{2n+1} = 2i \arctan 2,$$
$$\lim_{A \to \infty} I_A = 2i \arctan 2 - \pi i.$$

In particular the conjectured sum limit does not exist. Even modulo $2\pi i$, its subsequential limits disagree with the integral limit.

Proof. Let $u = 2m + 1/2$. Gamma recurrence, valid here for every $A \geq 1$, gives

$$R_A(m) = \frac{(A+iu)(A+1+iu)}{(A-iu)(A+1-iu)} = e^{i\phi_A(u)}, \qquad \phi_A(u) = 2\arctan\frac{u}{A} + 2\arctan\frac{u}{A+1}. \tag{9}$$

This is the lift (1) for $v = 1$, since it is continuous in u and vanishes at zero. The phase is odd, strictly increasing, and takes values in $(-2\pi, 2\pi)$. Its two crossings of the principal cut occur at $u = \pm h_A$, where $h_A = \sqrt{A(A+1)}$: indeed the two positive arctangents sum to $\pi/2$ exactly when their arguments have product one. Away from these crossings,

$$\operatorname{Log} R_A(m) = i\phi_A(u) - 2\pi i\,\mathbf{1}_{u>h_A} + 2\pi i\,\mathbf{1}_{u<-h_A}. \tag{10}$$

No integer sample lies on the cut: u is a half-integer whereas h_A^2 is an integer. The two cut values do not affect the integral.

Since $A < h_A < A + 1/2$, the cut locations in the m variable,

$$r_A = (h_A - 1/2)/2, \qquad l_A = (-h_A - 1/2)/2,$$

belong to $(-A, A)$ for $A \geq 1$ and satisfy $r_A + l_A = -1/2$. For $A = 2n$, the positive cut has $r_A \in (n - 1/4, n)$ and the negative cut has $l_A \in (-n - 1/2, -n - 1/4)$. There are $n + 1$ integers above r_A and n below l_A in $[-A, A]$. For $A = 2n + 1$, the corresponding intervals are $(n + 1/4, n + 1/2)$ and $(-n - 1, -n - 3/4)$; both counts are $n + 1$. Writing

$$U_A = \sum_{m=-A}^{A} i\phi_A(2m + 1/2), \qquad V_A = \int_{-A}^{A} i\phi_A(2m + 1/2)\,dm,$$

equation (10) therefore gives the exact identities

$$S_A = \begin{cases} U_A - 2\pi i, & A \text{ even}, \\ U_A, & A \text{ odd}, \end{cases} \qquad I_A = V_A - 2\pi i\big((A - r_A) - (l_A + A)\big) = V_A - \pi i.$$

Theorem 1, with $(b, c, v) = (1/2, 2, 1)$, gives $U_A, V_A \to 2i\arctan 2$, proving every asserted limit. □

Corollary 1 (A sufficient principal-branch range)**.** *If $4|y/q_y|\arctan(q_x/q_y) < \pi$, the original principal-log sum and integral both converge to the value in* (2)*. In particular this holds throughout $|y| \leq q_y/2$.*

Proof. Equations (6) and (8) give, uniformly for $|u| \leq cA + |b|$,

$$|L_A(u)| \leq 4|v|\arctan\frac{cA + |b|}{A + 1 - |v|} + \frac{2|v|}{A + 1 - |v|}.$$

Its limiting upper bound is strictly less than π under the hypothesis. For all sufficiently large A the lift is consequently the principal logarithm throughout the integration interval, and Theorem 1 applies. □

Scope and attribution. The continuous-log value above reproduces the gamma contribution used in the source (multiplication by $i/2$ gives its $-4xy\arctan(q_x/q_y)/(q_xq_y)$). It is not a claim about all later physical conclusions. The current six-page arXiv version and the six-page journal final, *Physical Review B* **98** (2018), 104511, present the phase formula without the conjecture's proof. The author's thesis, *Geometric phases in topological superconductors* (2018), Appendix C, p. 115 (PDF p. 125), still assumes the equality. Complete original and current preprints, the complete embedded journal final, and the complete relevant thesis appendix were read. No earlier full target claim was found in the bounded checks documented alongside this proof. These are authoring-agent checks, not independent peer review or formal certification. One problem is credited; the following are proposed research directions, not additional solutions.

Task 2: Classify principal-log convergence

Determine necessary and sufficient conditions on (b, c, v) for the principal-log sum to converge, accounting for every cut crossing and for arithmetic properties of the moving cut locations.

Task 3: Determine the full error expansion

Find successive powers of A^{-1} in the continuous-log sum and integral, with uniform remainder bounds on compact parameter sets, and determine when the leading discrepancy vanishes.

Task 4: Treat oblique complex sampling

Extend the continuous-log theorem to the complex lattice directions used for an oblique torus, specifying domains which avoid gamma poles and proving uniform control as directions approach a boundary.

Task 5: Track the phase through the original derivation

Reconstruct the full vortex-phase calculation with explicitly chosen logarithm lifts and cut corrections. Determine precisely which equalities hold in the complex plane and which only in the phase quotient, including their behavior across fundamental cells.